

Subject Description Form

Subject Code	ITC6863																																												
Subject Title	Technology IV: Smart Textiles & Clothing																																												
Credit Value	3																																												
Level	6																																												
Pre-requisite / Co-requisite / Exclusion	Nil																																												
Objectives	This subject will provide a guide for research students to carry out in-depth study in smart technologies for textiles and clothing.																																												
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Chart the development roadmap of the field b. Understand scientific principles of the smart textiles c. Use analytic method and characterization techniques for smart textiles d. Envisage applications of smart textiles and clothing																																												
Subject Synopsis/ Indicative Syllabus	This subject will provide a guide for research students to carry out in-depth study in smart technology for textiles and clothing, including principles, sensors and actuators, signal transmission and processing, multi-functional design, integration processes as well as analysis and evaluation.																																												
Teaching/Learning Methodology	Class studies, laboratories and self-studies will be conducted on the field and specific topics, e.g. thermal sensitive textiles and clothing, conductive fibers, shape memory alloy and polymers, photonic fibers and fabrics, wearable electronics and photonics. Group discussions and presentation will be conducted.																																												
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th></th><th></th></tr> </thead> <tbody> <tr> <td>1. continuous assessment</td><td>40%</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td></td><td></td><td></td><td></td></tr> <tr> <td>2. exam</td><td>60%</td><td></td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td></td><td></td></tr> <tr> <td>Total</td><td>100%</td><td colspan="6"></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d			1. continuous assessment	40%	✓	✓					2. exam	60%		✓	✓	✓			Total	100%						
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Student Study Effort Expected	Class contact:	
	▪ Lecture	5 Hrs.
	▪ Presentation/discussions	16 Hrs.
	▪ Laboratory demonstration	5 Hrs.
	Other student study effort:	
	▪ Literature studies	44 Hrs.
	▪ projects	50 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	<u>Monographs:</u> 1. Tao, X.M., 2001. <i>Smart fibres, fabrics and clothing</i>. Cambridge: Woodhead Publishing Co, 2. Tao X.M., 2005. <i>Wearable electronics and photonics</i>. Cambridge: Woodhead Publishing Co. 3. Mattila H.R. 2006. <i>Intelligent Textiles and Clothing</i>, Cambridge: Woodhead. 4. Seymour, S., 2010. <i>Functional aesthetics: visions in fashionable technology</i>. New York. Springer. 5. Hu, J.L., 2011. <i>Adaptive and Functional Polymers, Textiles and Their Applications</i>, London: Imperial College Press. <u>Journals:</u> 1. Science, 2. Natural Materials 3. Advanced Materials 4. Advanced Functional Materials 5. Chemistry of Materials 6. Applied Physics Letters 7. Textile Research Journal	